Just lab practice

Implementing vanilla RNN: Scratch:

$$h_t = \tanh(W_{xh} x_t + W_{hh} h_t + b_h)$$

$$y_t = W_{hy} h_t + b_y$$

$$P_t = \text{softmax}(y_t)$$

Instead of binary or like switch I. Loss function: Cross Entropy loss.

(I am just using sigmoid (It will be smooth switch)

At time t: $L_t = -\log(P_t[\text{target}])$

$$L = -\sum_t \log(P_t[\text{target}_t])$$

$$\frac{\partial L}{\partial y_t} = P_t - y_t^{\text{True}}$$

What happens in forward pass:

At time t: $h_t = \tanh(W_{xh}$

$$x_t = \begin{bmatrix} 0.5 \\ -0.1 \end{bmatrix} \quad h_{t-1} = \begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix} \quad C_{t-1} = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$$

$$W_{xi} = \begin{bmatrix} 0.5 & -0.3 \\ 0.4 & 0.1 \end{bmatrix}_{2\times2} \quad W_{hi} = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.3 \end{bmatrix}_{2\times2}$$

$$W_{xf} = \begin{bmatrix} -0.4 & 0.2 \\ 0.3 & 0.3 \end{bmatrix}_{2\times2} \quad W_{hf} = \begin{bmatrix} 0.05 & -0.1 \\ 0.2 & 0.1 \end{bmatrix}_{2\times2}$$

$$W_{xo} = \begin{bmatrix} 0.3 & 0.25 \\ -0.2 & 0.2 \end{bmatrix}_{2\times2} \quad W_{ho} = \begin{bmatrix} 0.15 & 0.05 \\ 0.1 & -0.2 \end{bmatrix}_{2\times2}$$

$$W_{xg} = \begin{bmatrix} -0.5 & 0.4 \\ 0.2 & -0.3 \end{bmatrix}_{2\times2} \quad W_{hg} = \begin{bmatrix} 0.2 & 0.1 \\ -0.1 & 0.05 \end{bmatrix}_{2\times2}$$

$C_t = ?$

$h_t = ?$

$$C_t = \underset{①}{f_t} \odot \underset{②}{C_{t-1}} + \underset{③}{i_t} \odot \underset{④}{g_t}$$

$$f_t = \sigma(W_{hf} h_{t-1} + W_{xf} x_t)$$

$$W_{hf} h_{t-1} = \begin{bmatrix} 0.05 & -0.1 \\ 0.2 & 0.1 \end{bmatrix}_{2\times2} \begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix}_{2\times1}$$

$$= \begin{bmatrix} 0 - 0.01 \\ 0 + 0.01 \end{bmatrix} = \begin{bmatrix} -0.01 \\ 0.01 \end{bmatrix}$$

$$W_{xf} x_t = \begin{bmatrix} -0.4 & 0.2 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.2 - 0.02 \\ 0.15 - 0.03 \end{bmatrix} = \begin{bmatrix} -0.22 \\ 0.12 \end{bmatrix}$$

$$f_t = \sigma\left(\begin{bmatrix} -0.01 \\ 0.01 \end{bmatrix} + \begin{bmatrix} -0.22 \\ 0.12 \end{bmatrix}\right) = \sigma\begin{pmatrix} -0.23 \\ 0.13 \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.5 \end{pmatrix}$$

$$=$$

$$C_{t-1} = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$$

$$i_t = \sigma\left(W_{hi} h_{t-1} + W_{xi} x_t\right)$$

$$= \sigma\left(\begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.3 \end{bmatrix}\begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.5 & -0.3 \\ 0.4 & 0.1 \end{bmatrix}\begin{bmatrix} 0.5 \\ -0.1 \end{bmatrix}\right)$$

$$= \sigma\left[\begin{pmatrix} 0+0.02 \\ 0+0.03 \end{pmatrix} + \begin{pmatrix} 0.25+0.03 \\ 0.2-0.01 \end{pmatrix}\right] = \sigma\left[\begin{pmatrix} 0.02 \\ 0.03 \end{pmatrix} + \begin{pmatrix} 0.28 \\ 0.19 \end{pmatrix}\right.$$

$$= \sigma \begin{bmatrix} 0.3 \\ 0.22 \end{bmatrix}$$

$$= \begin{pmatrix} \frac{1}{1.74} \\ \frac{1}{1.802} \end{pmatrix} = \begin{bmatrix} 0.574 \\ 0.554 \end{bmatrix}$$

$$g_t = \tanh\left(W_{hg} * h_{t-1} + W_{xg} x_t\right)$$

$$= \tanh\left(\begin{bmatrix} 0.2 & 0.1 \\ -0.1 & 0.05 \end{bmatrix}\begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix} + \begin{bmatrix} -0.5 & 0.4 \\ 0.2 & -0.3 \end{bmatrix}\begin{bmatrix} 0.5 \\ -0.1 \end{bmatrix}\right)$$

$$= \tanh\left(\begin{bmatrix} 0+0.01 \\ 0+0.005 \end{bmatrix} + \begin{bmatrix} -0.25-0.04 \\ 0.1+0.03 \end{bmatrix}\right)$$

$$= \tanh\left(\begin{bmatrix} 0.01 \\ 0.005 \end{bmatrix} + \begin{bmatrix} -0.29 \\ 0.13 \end{bmatrix}\right) = \tanh\begin{pmatrix} -0.28 \\ 0.135 \end{pmatrix}$$

$$= \begin{bmatrix} -0.2729 \\ 0.1342 \end{bmatrix}$$

$$c_t = \begin{pmatrix} 0.44 \\ 0.532 \end{pmatrix} \odot \begin{pmatrix} 0.2 \\ -0.2 \end{pmatrix} + \begin{pmatrix} 0.574 \\ 0.554 \end{pmatrix} \cdot \begin{pmatrix} -0.2729 \\ 0.1342 \end{pmatrix}$$

$$= \begin{pmatrix} 0.088 \\ -0.1064 \end{pmatrix} + \begin{pmatrix} -0.1566446 \\ 0.0743468 \end{pmatrix} = \begin{pmatrix} -0.0686 \\ -0.032 \end{pmatrix}$$

$$h_t = O_t \odot \tanh(c_t)$$

$$O_t = \sigma\left(W_{ho} h_{t-1} + W_{xo} x_t\right)$$

$$= \sigma\left(\begin{bmatrix} 0.15 & 0.05 \\ 0.1 & -0.2 \end{bmatrix}\begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.3 & 0.25 \\ -0.2 & 0.2 \end{bmatrix}\begin{bmatrix} 0.5 \\ -0.1 \end{bmatrix}\right)$$

$$= \sigma\left[\begin{pmatrix} 0 + 0.005 \\ 0 - 0.02 \end{pmatrix} + \begin{pmatrix} 0.15 - 0.025 \\ -0.1 - 0.02 \end{pmatrix}\right]$$

$$= \sigma\left[\begin{pmatrix} 0.005 \\ -0.02 \end{pmatrix} + \begin{pmatrix} 0.125 \\ -0.12 \end{pmatrix}\right] = \sigma\begin{pmatrix} 0.13 \\ -0.14 \end{pmatrix}$$

$$\sigma\begin{pmatrix} 0.13 \\ -0.14 \end{pmatrix} = \begin{pmatrix} \frac{1}{1.878} \\ \frac{1}{2.15} \end{pmatrix} = \begin{pmatrix} 0.532 \\ 0.465 \end{pmatrix}$$

$$\tanh(c_t) = \begin{pmatrix} \tanh 2 \\ -0.0686 \\ -0.032 \end{pmatrix} = \begin{matrix} -0.06849 \\ -0.03199 \end{matrix}$$

$$h_t = \begin{pmatrix} 0.532 \\ 0.465 \end{pmatrix}\begin{pmatrix} -0.06849 \\ -0.03199 \end{pmatrix}$$

$$= \begin{pmatrix} -0.036 \\ -0.014 \end{pmatrix}$$